Neither conduction nor a fixed delivered energy fraction reproduces sixteen two-cell propagation tests: a pre-registered kinetic study

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Abstract

[abstract, written after Stage 11]

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1. Introduction

Thermal runaway in one cell of a pack can spread to its neighbours, and models of the pack are used to judge whether and when it does [1–4]. They represent each cell as one temperature or resolve it, and let a neighbour run away when its temperature reaches an onset value or when Arrhenius heat sources outrun its losses. Their heat paths are conduction through contacts, tabs and holders and radiation between the cans [3], and some represent what the failing cell vents: the share of its energy carried off with the ejecta [2], the vent gas and its reactions [5], or the gas layer that forms between the roll and the can [4]. Propagation tests in modules of pouch and cylindrical cells report the order and timing of failures in a few configurations [6–8], and models have been compared with such tests one configuration at a time [1, 4].

Calorimetry has shown how much of a failing cell's energy does not stay in it. Fractional thermal runaway calorimetry of 18650 cells finds that typically 20 to 30 % of the total energy leaves through the casing and the rest with the ejecta [9], and the Battery Failure Databank collects such measurements for many cells, the LG M50 among them [10]. A model that heats a neighbour by conduction and radiation from the failing cell's body therefore omits most

of the energy, and a model that represents the ejecta must choose how much of them reaches the neighbour.

The tests that could check that choice report little per test. Fransson et al. [11] ran sixteen two-cell propagation tests of LG M50 cells in direct contact, joined by welded tabs, eight in series and eight in parallel; for each they report whether the neighbour propagated, when, and the two cells' maximum surface temperatures, from one thermocouple per cell. Nine propagated, at 147 to 342 s, and the paper infers from its records that propagation needed at least 150 °C in the neighbour. Read, Jones and Marco [12] report tests of a seven-cell module of the same cell, in which propagation followed the trigger's sidewall rupture. Such datasets allow a comparison test by test, which the timing of one configuration cannot give.

The chemistry that turns heating into runaway is a network of decomposition reactions of the SEI, the electrolyte, the electrodes and the separator [13]. Koenig, Zhao and Deng [14] inferred such a network from calorimetry with a Bayesian chemical reaction neural network and published the posterior over its parameters. That makes it possible to ask how much of a model's prediction the kinetic uncertainty could move, beside the uncertainty of the thermal properties [15, 16]. A published two-cell study [17] solved the neighbour of a failing LG M50 cell both as one temperature and as a resolved field, with identical inputs, and compared both with fixed onset temperatures; its geometry and solver are this study's starting point.

Three questions have not been put to a model against a set of tests. Do a resolved and a lumped neighbour, given the same kinetics, agree on when the chemistry starts, and what sets it? Do conduction and radiation alone account for the measured outcomes, and if not, what must be delivered, and how much? And does the criterion that decides ignition, local at the hottest point or the whole cell's, predict which tests propagate and when? Answering them needs predictions fixed before the runs, because a model with a free delivered energy can be made to reproduce any one test.

This paper reports a study that put these questions to the model in eleven pre-registered stages. Its contributions are:

- the decomposition kinetics integrated in a resolved and a lumped neighbour, over 200 draws of their posterior and 30 draws of the roll's published thermal properties, with the finding that the two neighbours disagree on the onset and that the kinetics set both;
- a test of conduction and radiation, and of seven ignition criteria, against

the sixteen measured tests, test by test;

- the inverse reading: the fraction of the trigger's ejected energy each test needs, read from its outcome, its time and its peak;
- the one published neighbour history set against those readings, and held-out scores of every predictor against one probability for every test;
- bands of the central numbers under kinetic and thermal draws chosen by rank before the runs;
- a pre-registration record: 70 predictions committed before their runs, 36 of which failed, every verdict reported (Table 4).

2. Methods

The study ran in eleven stages. Each stage's predictions, with the expectation held for each, were committed to a public repository and pushed before any of its runs, and the stage's inputs were recorded in a decisions log before they were used; Section 2.10 gives the record. The methods below are those of the stages whose results this paper reports, in the order of the claims.

2.1. Two cells, two neighbours

The geometry is that of a published two-cell study [17]: two LG M50 cylindrical cells (21700 format, 21.1 mm in diameter and 70.1 mm long) side by side with parallel axes (Fig. 1a). The trigger (cell 1) is one temperature, which releases its body energy E_{b} inside itself as a triangular pulse. The neighbour (cell 2) is solved twice from the same history of the trigger: as one temperature, the lumped neighbour, and as a three-dimensional field, the resolved neighbour, a half model on 19,355 nodes with its symmetry plane through both axes. The resolved roll conducts 1.3 $\mathrm{W\,m^{-1}\,K^{-1}}$ across its layers and 25 $\mathrm{W\,m^{-1}\,K^{-1}}$ along them, inside a 0.25 mm steel can; both neighbours hold the cell's heat capacity, 61.75 $\mathrm{J\,K^{-1}}$. In Stages 1 to 4 the cells stand 2 mm apart joined by busbars, as the two-cell study archived them: Stage 1 integrates the kinetics along its archived temperature histories, Stage 2 feeds their heat back into both solvers (one-way), and Stages 3 and 4 repeat Stage 1 over the posterior. From Stage 5 on, the cans touch along a line, as in the tests [11]. Within 45° of the line of contact the conductance

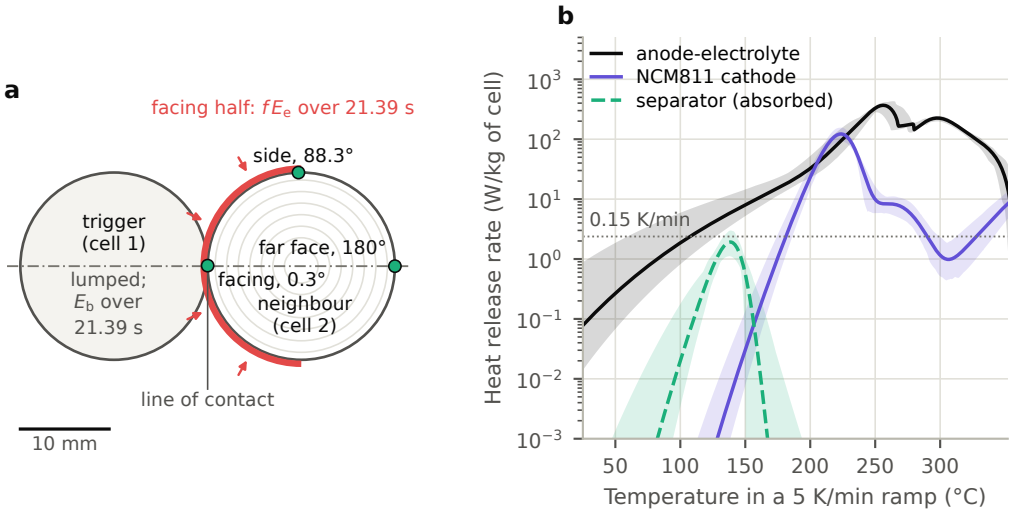


Figure 1: The two cells and the kinetics. (a) Cross-section at mid-length. The trigger (cell 1, one temperature) releases its body energy E_b over 21.39 s; the neighbour (cell 2) is solved as one temperature and as a resolved half model (symmetry plane dash-dotted), the cans touching along the line of contact. From Stage 8 a fraction f of the ejected energy E_e enters through the half of the neighbour's can facing the trigger (red). The three nodes at mid-length bracket the thermocouple's unreported angle. (b) Heat release rate of each submodel of the kinetics in a 5 K min^{-1} ramp from 25 °C, per kilogram of cell: the posterior mean (lines) and the 2.5 to 97.5 % band over the 200 posterior draws; the separator's melting absorbs heat, and its magnitude is shown. Dotted: a self-heating rate of 0.15 K min^{-1} at the cell's specific heat. The schematic and every figure were drawn by scripts, written with a generative AI model (Section 2.10), from the results files.

per unit area is a steel-on-steel contact conductance h_c in series with the air wedge, $h(\theta) = k_{air}/(\delta(\theta) + k_{air}/h_c)$ with $\delta = 2R(1 - \cos\theta)$, and h_c is 200, 2,000 or 3,700 W m^{-2} K^{-1} for the low, central and high contact [18]; nickel-strip tabs join the cells (5, 6.8 and 10 mW K^{-1} each, one in series and two in parallel) [17]; the cans radiate to each other (view factor 0.1817 [19]) and to the room, and lose heat to it by convection, at 25 °C. The hot spot is the resolved neighbour's hottest node at each time; the far face is the surface node at mid-length opposite the line of contact. A step splits the reaction, integrated node by node, from the backward-Euler conduction step, at a length that keeps any node's reaction heating within 1 K a step. Table 1 lists the parameters and their sources.

Table 1: The model's parameters and their sources: a published source, the two-cell study whose geometry and solver this study takes [17], or an entry of this study's decisions log (D, in the repository [20]). Values in parentheses are brackets run as sensitivities or the ranges of the thermal draws.

Quantity	Value	Source
Geometry		
Cell diameter and length	21.1 and 70.1 mm	two-cell study [17]
Can: wall, conductivity; mass and specific heat	0.25 mm, 15 $\mathrm{W\,m^{-1}\,K^{-1}}$; 9 g, 500 $\mathrm{J\,kg^{-1}\,K^{-1}}$	two-cell study [17]
Resolved neighbour	half model, 19,355 nodes (145,195 halved)	this study (D42)
Thermal properties		
Roll conductivity, across and along the layers	1.3 and 25 $\mathrm{W\,m^{-1}\,K^{-1}}$ (draws 0.998 to 1.38; 24.9 to 25.8)	[15, 16]; D27
Cell heat capacity	61.75 $\mathrm{J\,K^{-1}}$ (65 g at 950 $\mathrm{J\,kg^{-1}\,K^{-1}}$; roll draws 842 to 1110 $\mathrm{J\,kg^{-1}\,K^{-1}}$)	two-cell study [17]; [15]; D27
Paths		
Contact conductance between the cans, h_c	200, 2,000, 3,700 $\mathrm{W\,m^{-2}\,K^{-1}}$	[18]; D36
Air in the wedge	0.03 $\mathrm{W\,m^{-1}\,K^{-1}}$	two-cell study [17]
Tabs (series one, parallel two)	5, 6.8, 10 $\mathrm{mW\,K^{-1}}$ per tab	two-cell study [17]; D37
View factor between the cans	0.1817	[19]; D38
Emissivity; room convection; ambient	0.8; 10 $\mathrm{W\,m^{-2}\,K^{-1}}$; 25°C	two-cell study [17]; D38
Trigger		
Model and release	one temperature; triangular over 21.39 s (17.53, 25.24 s)	two-cell study [17]; supplement of [11] with a 25°C start; D52
Body energy E_b, one per test	9.0 to 54.3 kJ	Tables 3, 4 of [11]; D65
Stage 5 energy; with the ejecta (Stage 6)	33.2 kJ; 107.0 kJ	[10]; D38, D53
Kinetics		
Model and posterior	B-CRNN, mean and 200 joint draws	[14]; D1, D3, D22
Reactant masses	per 720 g reference cell	[14]; D4
Initial SEI amount	0.15 (fresh); 0 (aged)	[14]; D66
Separator's melting heat	absorbed	[14], Fig. 3(e); D5
Levels		
Propagation (C1, C2)	255.7°C within 900 s	internal-short band of [14]; D35
The paper's criterion (C3)	150°C	[11]
Separator's melting (C4, C5, C7)	138.5°C	[14]; D5, D71
Self-heating onset	0.15 $\mathrm{K\,min^{-1}}$	[14]
Event (Stage 7)		
Stored energy released from the separator's melting	65.52 kJ (63.36 kJ) over 21.39 s	cell's rating; D68
Delivery (Stage 8 on)		
Ejected-to-body ratio r	3.105 (2.218 to 4.156)	[10]; D77
Shape, duration, region	triangular, 21.39 s (10, 60 s), facing half (cap-end quarter)	D77
Joule heat in the parallel tests	5 and 20 % of 65.52 kJ	D80
Gas layer, largest resistance	0.03077 (facing half), 0.00962 $\mathrm{m^2\,K\,W^{-1}}$	[4]; D82
Nodes		
Thermocouple bracket at mid-length	0.33°, 88.28°, 180°	D102

2.2. The kinetics, their posterior and the SEI

The decomposition kinetics are the Bayesian chemical reaction neural network of Koenig, Zhao and Deng [14], trained on calorimetry of NCM811 and graphite: seven anode, electrolyte, SEI and binder reactions, three cathode reactions and the separator's melting, each an Arrhenius rate with learned orders, whose heat enters per unit cell mass at the reactant mass fractions of the source's reference cell. The model is the mean of each parameter over the source's posterior particles; its DSC curves reproduce the source's at 5 $\mathrm{K\,min^{-1}}$, and Fig. 1b shows each submodel's heat release in such a ramp. The posterior enters as 200 joint draws, each one particle of each submodel's file taken independently with replacement. The SEI starts fresh, at the source's initial amount; the aged SEI of Stage 7 starts at none. A node starts to self-heat when its reaction heat alone heats it at 0.15 $\mathrm{K\,min^{-1}}$, and it propagates in the model when it reaches 255.7 °C, the lower edge of the source's internal-short band [14]. An emulation of the source's accelerating-rate test gives a runaway onset of 179.3 °C, inside the source's band of 169.0 to 198.0 °C; its internal-short temperature at the stated trigger rate is 199.8 °C, below the source's band, a mismatch this study does not resolve.

2.3. The tests, the triggers and the ejected energy

Fransson et al. [11] ran sixteen two-cell propagation tests of LG M50 cells in direct contact, joined by welded tabs, eight in parallel and eight in series, the trigger penetrated by a nail and a thermocouple fastened at the middle of each cell's surface (Table 2; tests are named here without the prefix M50). Nine propagated, at 147 to 342 s from the trigger's runaway. The neighbours of five of the seven that did not peaked at 71 to 133 °C; the other two report 626 and 808 °C, which may not be the neighbour's surface, and are read as exploratory. The model's clock starts with the trigger's release, read as the start of its runaway. Each test gets its own trigger: the body energy E_b that puts cell 1's peak at the test's measured trigger maximum at the central contact, released over 21.39 s, the mean of the two trigger rise times the paper's supplement gives [11]. E_b runs from 9.0 kJ (PT_Exp1, 164 °C) to 54.3 kJ (SP_Exp3, 832 °C). The energy a trigger ejects as vent gas, ejecta and flame is $E_\mathrm{e} = r\,E_\mathrm{b}$, with $r = 3.105$ the median ejected-to-body ratio of the ten LG M50 nail tests of the Battery Failure Databank [10], bracketed by their smallest and largest, 2.218 and 4.156; E_e runs from 27.9 to 168.6 kJ.

Table 2: The sixteen two-cell LG M50 tests of Fransson et al. [11] (Tables 3 and 4 of the paper), with the trigger's body energy E_b that puts the model's trigger at the measured peak at the central contact over 21.39 s, and the ejected energy $E_e = 3.105\,E_b$. Times run from the trigger's runaway to the neighbour's onset. The two maxima marked [a] may not be the neighbour's surface and are read as exploratory.

Test	Connection	Trigger peak (°C)	E_b (kJ)	E_e (kJ)	Propagated	Time (s)	Neighbour max. (°C)
PT_Exp1	parallel	164	9.0	27.9	no		71
PT_Exp2	parallel	679	43.4	134.8	no		808[a]
PT_Exp3	parallel	507	31.6	98.1	no		118
PT_Exp4	parallel	630	40.0	124.2	yes	272	607
PP_Exp1	parallel	787	51.1	158.6	yes	152	381
PP_Exp2	parallel	644	41.0	127.2	yes	147	699
PP_Exp3	parallel	686	43.9	136.3	no		105
PP_Exp4	parallel	728	46.9	145.5	yes	275	523
ST_Exp1	series	788	51.1	158.7	no		128
ST_Exp2	series	723	46.5	144.3	no		133
ST_Exp3	series	282	16.6	51.7	yes	177	763
ST_Exp4	series	756	48.8	151.6	yes	342	733
SP_Exp1	series	728	46.8	145.4	yes	223	546
SP_Exp2	series	755	48.7	151.3	yes	149	698
SP_Exp3	series	832	54.3	168.6	yes	210	420
SP_Exp4	series	625	39.6	123.0	no		626[a]

2.4. The criteria

Stage 7 read seven candidate ignition criteria from each run with no new physics: the hot spot reaches 255.7 °C (C1); the lumped neighbour reaches 255.7 °C (C2); the lumped neighbour reaches 150 °C, the paper's criterion (C3); the separator's melting, 138.5 °C, at the hot spot (C4) or in the lumped neighbour (C5); and C1 and C4 with the aged SEI (C6, C7). A criterion predicts propagation when it is met within 900 s, and its time when it is met. From Stage 8 on, C1 is the local criterion and C2 the whole-cell clock. Stage 7 also ran the internal short as an event: the neighbour's stored 65.52 kJ released over 21.39 s from its separator's melting, a modelling choice.

2.5. The delivery and its brackets

From Stage 8 on, a fraction f of the trigger's ejected energy reaches the neighbour as a triangular pulse over 21.39 s from the start of the release, a uniform flux through the half of the can facing the trigger (Fig. 1a);

the lumped neighbour receives the same energy. f is a model quantity, the fraction a conduction-and-delivery model needs, not a measurement of what reached the neighbour. The brackets, run as sensitivities, are a delivery over 10 or 60 s, the quarter of the facing half nearest the cap, the ratio's two ends, Joule heat of 5 or 20 % of the stored energy in the parallel tests, the gas layer between the roll and the can of Hoelle, Zimmermann and Hinrichsen [4], and, in Stage 9, a constant power over 120, 240 or 360 s.

2.6. The readings: f^*, f_t and f_max

f^*, a test's threshold for a criterion, is the bracket $(\mathrm{lo}, \mathrm{hi}]$ of the smallest fraction at which the model meets it, from $f = 0$ through 0.1, 0.2 and 0.3, then 0.5 and 1, halving to a width of at most 0.02; it is 0 where the run with no delivery already meets it, and its value is the bracket's midpoint. A propagated test's outcome is reproduced at f^* and above, a non-propagated test's below it. Stage 9 read each test's own measurement as a fraction: f_t, at which a criterion is met at the measured time, for the propagated tests (bisection to 0.005, or a run within 10 s), and f_max, at which the far face peaks at the measured maximum, for the others (bisection to 0.005, or a peak within 5 K, over 900 s). A propagated test is read by its time only and a non-propagated test by its peak only, because after ignition the model's temperatures are not comparable with the measured maxima.

2.7. The history and the thermocouple's angle

No raw series of the tests is published. Stage 10 read the one published history of a neighbour before propagation, PT_Exp4's, and the one of a test that did not propagate, PP_Exp3's, out of the vector paths of the supplement's Figure 10 [11], twice by separate code, to 2.47 and 2.46 K and 0.45 s, and checked them against the page's pixels. Each model history at the far face was compared with PT_Exp4's record by the root mean square of the difference over 0 to 271.5 s, the onset the figure marks. The paper places the thermocouple at the middle of the surface at an angle it does not give, so three nodes at mid-length bracket it: facing the trigger (0.33° from the line of contact), at the side (88.28°) and at the far face (180°).

2.8. Held-out scores and ranks

Each predictor (f^*(C1), f^*(C2), and Stage 9's fraction, f_t(C2) or f_max) was fitted on the eight series tests and scored on the eight parallel ones, and the other way round: a penalised (Firth) logistic regression of the outcome

on the predictor, against one probability for every test, $(k+0.5)/(n+1)$, by the mean log score and the Brier score, with permutation p-values and bands from 10,000 relabellings of the outcomes. A predictor counted as better if it beat the constant by 0.1 in the mean log score, a margin registered before the scoring. Spearman's correlation ranked each quantity against the outcomes and the measured times.

2.9. The bands of Stage 11

Stage 11 asked how much of the central numbers the kinetic and the thermal uncertainty could account for. Four kinetic draws were chosen by rank, before its predictions, from Stage 3's 200 draws ranked by the hottest node's onset in the headline passive case (ranks 5, 50, 151 and 196), and four thermal draws from Stage 3's 30 Latin hypercube draws of the roll's published properties [15, 16]: the smallest and largest conductivity across the layers and the smallest and largest heat capacity. A thermal draw sets the neighbour's roll; the trigger's history stays central. For four tests (the smallest and largest trigger energy, and the two tied nearest the median) $f^{*}(\mathrm{C1})$ and $f^{*}(\mathrm{C2})$ were searched under each draw from Stage 8's bracket; the far face at $f = 0$ was run under each thermal draw and at the low and high contacts for the five low-maximum tests; and the nine propagated tests were run at their $f_{\mathrm{t}}(\mathrm{C2})$ under each kinetic draw. Its bands are over four draws a side, not over the posterior; the full-posterior quantity the study has is Stage 3's band of the onset lead over 200 draws.

2.10. Pre-registration and verification

Each stage's predictions file was committed and pushed before any of its runs; its commit, push time and the hash of its run list are in the repository's record [20], and the files were not changed afterwards. Through Stage 10 the study registered 64 predictions, of which 32 failed; Stage 11 registered six more (Table 4; every prediction, its expectation and its verdict in the supplement). Every stage closed the energy account of every run (the largest residual 2×10^{-11} of the heat taken in), halved the mesh spacing and the step limits on a run near its threshold (the hot spot's peak rise moved by at most 0.52 %), recomputed its registered numbers by separate code, and had every registered number and verdict read again, before its texts were written, by a check that had not drafted them. The study's code was written, its runs made and checked, and its texts drafted by a generative AI model, Claude Opus 5.5 (Anthropic), working stage by stage under the author's direction;

each stage's independent read was made by a separate instance of the same model that had not drafted the stage's texts. The predictions were committed before the runs whoever drafted them, and every result is reproducible from the archived code and records [20].

3. Results

3.1. The resolved and the lumped neighbour disagree on when the chemistry starts, and the kinetics set both

Every history that self-heats first does so at 108.4 °C, by decomposition of the fresh SEI (Fig. 1b). With copper busbars (500 mW K^{-1}) and the largest Databank energy, 33.2 kJ, the resolved neighbour's hottest node starts 23.4 and 26.4 s before the lumped node (10 and 30 s releases). Over the 200 posterior draws the lumped node never starts first in any of 42 cases, and at the headline case the lead runs from 15.4 to 58.9 s (2.5 to 97.5 %), with two modes (Fig. 2a). Over the published spread of the roll's conductivity and heat capacity the hottest node's onset varies 0.027 times as much as over the kinetic posterior, and the lumped node's 0.11 times: the kinetics set both onsets, and the anode-electrolyte submodel, which holds the SEI, carries 0.997 of the lead's variance. Fed back, the reaction heat moves the hot spot's entry into the internal-short band by less than 0.01 s: conduction from the busbar sets it.

3.2. With the trigger's body energy, conduction and radiation do not ignite the neighbour

With the largest body energy the Databank gives for this cell, 33.2 kJ, conduction and radiation alone take neither neighbour to 255.7 °C within 900 s in any of 5,248 runs with the cans in contact, across the contact, tab, energy and ambient brackets and the 200 draws; the hot spot peaks at most at 226.7 °C and the lumped neighbour at 180.6 °C. With the trigger's release matched to the tests' durations the trigger peaks at 526 to 537 °C, inside the measured 164 to 832 °C, and neither neighbour ignites in 442 runs with the draws (Fig. 3b). Only the Databank's total energy with the ejecta, 107 kJ, which puts the trigger at 1,424 to 1,494 °C, takes the hot spot to 255.7 °C, in every run at 8 to 17 s. By the paper's 150 °C criterion only the lumped neighbour at the central contact crosses inside the measured window (Fig. 3a). The measured trigger peaks reach 832 °C, and putting the model's

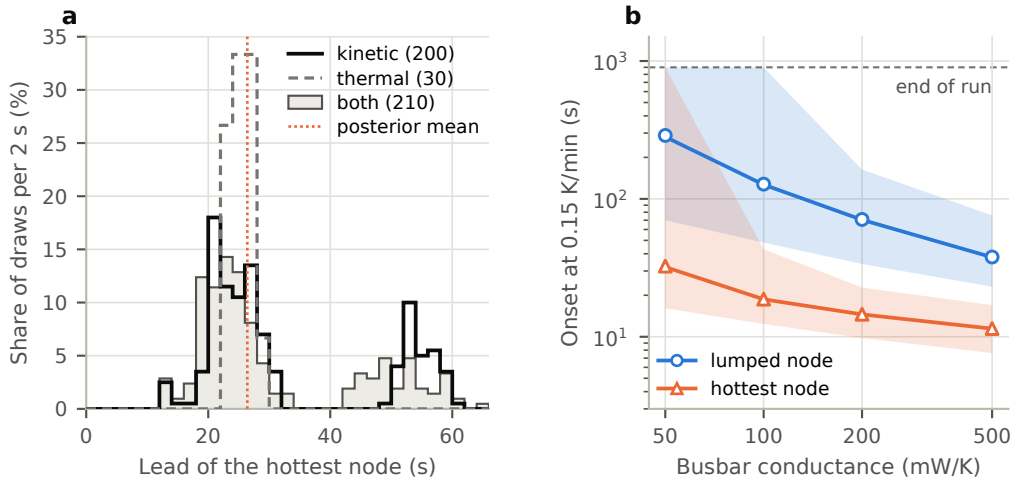


Figure 2: The onset in the published two-cell geometry (cells 2 mm apart, busbars), 33.2 kJ released over 30 s. (a) The hottest node's lead over the lumped node (the lumped node's onset at 0.15 K min^{-1} less the hottest node's) at 500 mW K^{-1}: over the 200 kinetic draws at the central thermal properties, the 30 thermal draws at the posterior-mean kinetics and the 210 combined pairs; dotted, the posterior mean. (b) Onset against the busbar conductance: median (lines) and 2.5 to 97.5 % band over the 200 kinetic draws, and the posterior mean (markers). A band that reaches the end of the run holds draws in which that node does not start within 900 s.

trigger at each test's own peak takes up to 54.3 kJ (Table 2); only with those larger triggers does conduction ignite the neighbour (Section 3.3).

3.3. The local criterion predicts which tests propagate, but never when; the whole-cell clock can match when, but rarely predicts whether

With each test's own trigger and no delivery, the hot spot at 255.7 °C (C1) predicts propagation in eight tests, six of them hits, and gets 11 of 16 outcomes right, more than any other criterion (Fig. 4); every hit comes 130 to 323 s before the measured time, because the hot spot crosses by 21.7 s. The lumped neighbour at 255.7 °C (C2) predicts two tests, both hits, 11.5 and 125.3 s after the measured times; at 150 °C (C3) and at the separator's melting (C4, C5) the criteria predict fourteen and get nine right. The aged SEI changes no hot-spot verdict. The internal short as an event overshoots: it takes the hot spot to 1,504 to 1,584 °C, about twice the hottest propagating neighbour, 763 °C. Near its own threshold the whole-cell clock has the induction the tests show and the hot spot none (Fig. 6a): with the

21.39 s delivery, C2 reproduces six of the nine measured times within 10 s, at f_{t}(C2) of 0.0016 to 0.047 (0.27 to 5.96 kJ), while C1, wherever it crosses, does so by 21.1 s and reproduces none.

3.4. A delivered part of the ejected energy is needed, and what each test needs varies 42- to 208-fold

For the hot spot, f^{*}(C1) is 0 in eight tests and at most 0.1625 in all; for the lumped neighbour, f^{*}(C2) is at most 0.05 in thirteen tests and at most 0.4625 in all, both highest in PT_Exp1, the smallest trigger (Fig. 5a, Table 3). The two groups overlap: the nine propagated tests need f^{*}(C1) of 0 to 0.0875 and the seven that did not reach it at 0 to 0.1625 (f^{*}(C2): 0 to 0.2375 against 0 to 0.4625), so no single fraction reproduces the sixteen outcomes. At a common fraction of 0.1, 0.2 or 0.3 every clock crosses by 88.3 s after the release and every hit comes 126 to 339 s early (Fig. 5b). Read from each test's own time or peak, the fraction the lumped neighbour's criterion needs varies 47.6-fold between tests with the 21.39 s delivery and 42.2- to 208.1-fold with the constant powers, and the propagated and non-propagated tests overlap in it too (Fig. 6b). Among the brackets, the ratio's ends scale every f^{*} by 0.75 to 1.40; with the gas layer of Hoelle et al. [4] the hot spot of the three tests nearest the median trigger stays below 255.7 °C at $f = 0.2$ (192.4 to 196.5 °C), where without it it crosses at 3.5 to 3.7 s.

3.5. Conduction alone overheats the far face in four of the five non-propagated tests

With no delivery the far face already peaks 13.8 to 56.3 K above the measured maxima of PT_Exp3, PP_Exp3, ST_Exp1 and ST_Exp2, so their $f_{\max}$ is 0; PT_Exp1, the smallest trigger, needs 0.075 (2.09 kJ) for its 71 °C. That 0.075 lies above the smallest f_{t}(C2), SP_Exp3's 0.0016, while the four zeros lie below every f_{t}: the fractions read from the times and the peaks do not separate the groups either. The thermocouple's angle does not remove the overshoot: at 88.3° or facing the trigger the four low maxima still need no delivery (overshoot 13.8 to 159.9 K), and only facing the trigger, with all five at 0, do the fractions separate the groups. In PP_Exp3's published record, the far face with no delivery lies 41.7 K above the measurement on average over 500 s (peaks 161.3 against 102.0 °C).

Table 3: The fractions of the ejected energy each test needs in the model, and Stage 11's ranges. (a) f^*(C1) and f^*(C2): the threshold bracket (lo, hi] at which the hot spot (C1) or the lumped neighbour (C2) reaches 255.7°C with the 21.39 s delivery, 0 where no delivery is needed. f_t(C2): the fraction at which the lumped neighbour's crossing comes at the measured time ([b] the search ended at the threshold, its bracket given). f_{max}: the fraction at which the far face peaks at the measured maximum ([a] exploratory). (b) Over the posterior mean and Stage 11's draws: f^*(C1) and f^*(C2) over the nine (four kinetic, four thermal, the mean), the far face's peak at $f = 0$ less the measured maximum over the central case, the four thermal draws and the two contact brackets, and the lumped neighbour's crossing at f_t(C2) under the four kinetic draws, with the number that do not cross within 900 s.

(a) Stages 8 and 9, at the posterior mean

Test	f^*(C1)	f^*(C2)	f_t(C2) or f_{max}
PT_Exp1	(0.15, 0.1625]	(0.45, 0.4625]	0.075
PT_Exp2	(0, 0.0125]	(0.0125, 0.025]	0.5625[a]
PT_Exp3	(0.025, 0.0375]	(0.075, 0.0875]	0
PT_Exp4	(0, 0.0125]	(0.0375, 0.05]	(0.0375, 0.040625][b]
PP_Exp1	0	0	0.01875
PP_Exp2	(0, 0.0125]	(0.025, 0.0375]	0.046875
PP_Exp3	(0, 0.0125]	(0.0125, 0.025]	0
PP_Exp4	0	(0, 0.0125]	0.0125
ST_Exp1	0	(0, 0.0125]	0
ST_Exp2	0	(0.0125, 0.025]	0
ST_Exp3	(0.075, 0.0875]	(0.225, 0.2375]	(0.23125, 0.234375][b]
ST_Exp4	0	(0, 0.0125]	(0.00625, 0.009375][b]
SP_Exp1	0	(0.0125, 0.025]	0.01875
SP_Exp2	0	(0, 0.0125]	0.025
SP_Exp3	0	0	0.0016
SP_Exp4	(0, 0.0125]	(0.0375, 0.05]	0.265625[a]

(b) Stage 11: ranges over the posterior mean and the draws

Test	f^*(C1)	f^*(C2)	Far face at $f = 0$ (K)	C2 crossing, kinetic draws (s)
PT_Exp1	0.14375–0.15625	0.39375–0.49375	−20.9 to −8.4	
PT_Exp3			−17.6 to 21.3	
PT_Exp4				191–197 (2 none)
PP_Exp1				140–150
PP_Exp2				138–155
PP_Exp3	0–0.00625	0.00625–0.03125	16.6 to 66.7	
PP_Exp4				230–237 (2 none)
ST_Exp1			−1.0 to 57.3	
ST_Exp2	0	0–0.01875	−12.8 to 41.2	
ST_Exp3				77–80 (2 none)
ST_Exp4				233–240 (2 none)
SP_Exp1				201–275
SP_Exp2				144–157
SP_Exp3	0	0–0.00625		198–230

3.6. The one published neighbour history follows a short delivery near its threshold, and no long one

PT_Exp4's thermocouple reaches 141.0 °C by 45 s and rises steadily to 197.3 °C at the onset, 271.5 s (Fig. 7a). At the far face, the 21.39 s delivery at the two ends of its C2 bracket ($f = 0.0375$ and 0.040625) lies 16.3 and 16.8 K from the record (root mean square to the onset), the constant powers at their f_{t} 36.7 to 37.2 K, and no delivery 47.4 K. The difference comes from the first 100 s, the record's rise to 141 °C by 45 s, which only the short delivery follows; after 100 s the short and the long C2 readings lie 8.8 to 12.0 K from the record, and none has its last rise. The record's first 8.5 s, a spike to 102 °C, is read as the thermocouple's exposure at the trigger's runaway rather than the can's temperature; it holds 41.2 % of the short delivery's squared difference. One test's history decides between the readings for that test only.

3.7. Held out, nothing the model yields beats one probability for every test

Fitted on one connection and scored on the other, no predictor beats one probability for every test by the registered margin of 0.1 in the mean log score; the largest gain is 0.062, Stage 9's fraction fitted on the series tests, and the permutation p-values run from 0.23 to 0.75 (Brier score, 0.20 to 0.75) (Fig. 7b). No model quantity ranks with the outcomes or the times (p at least 0.13), while f^{*}(C1) and f^{*}(C2) rank with the trigger's energy ($\rho = -0.92$ and -0.97): the thresholds mostly restate the trigger. The sixteen tests cannot adjudicate between these models.

3.8. How much of this the kinetic and thermal uncertainty could account for

[Stage 11 paragraph, written after its runs and the independent read]

3.9. The predictions that failed, and what each failure taught

Of the 70 predictions, 36 failed (Table 4). Twenty-two of the failures were foreseen in their predictions files, mostly because an earlier stage's records already fixed them, and are reported as failures all the same; the others each taught something. In Stage 2 the reaction heat was expected to push the hottest region into the internal-short band at 200 mW K^{-1}; it added 0.6 and 0.7 K, so the hot spot's peak is set by conduction, not by its chemistry. In Stage 3 the roll's thermal properties were expected to set the hottest node's onset more than the kinetics; they move it 0.027 times as much, and the kinetics set both onsets. In Stage 4 a tighter posterior of the SEI submodel was expected to narrow the lead by over 40 %; it narrows it by

Table 4: The predictions by stage: committed before the stage's runs, held, failed, and of the failures those foreseen, which the predictions file had recorded before the runs as expected to fail, as already fixed by earlier records or an energy balance, or as failing under a condition it named that then held. The stages' own texts label 17 of the 21 through Stage 10 so (Supplement S2). Every prediction, its expectation and its verdict are in the supplement.

Stage	Committed	Held	Failed	Foreseen
1 and 2: passive and reactive onset	7	6	1	0
3: onset over the posterior and thermal draws	6	3	3	2
4: which submodel sets the lead	5	4	1	0
5: neighbours against the tests	7	1	6	0
6: the paper's 150°C criterion; matched trigger	7	1	6	6
7: seven criteria, test by test	8	3	5	5
8: the delivered fraction f^*	8	5	3	3
9: fractions from times and peaks	8	5	3	3
10: one history; held-out scores	8	4	4	2
Stages 1 to 10	64	32	32	21
11: bands under the draws	6	2	4	1

14.6 %, and an exploratory analysis found that the lead has two modes, which a tighter posterior keeps. In Stage 5 the resolved neighbour was expected to propagate inside the measured window and in 30 to 90 % of the factorial runs; it propagated in none, the four predictions about a propagation's time, surface temperature and heating rate failed with it, and that is what led to the delivered energy. In Stage 10 the long deliveries were expected to follow PT_Exp4's history better than the short one; the short one followed it, because only it reproduces the rise to 141 °C by 45 s; and the survey found a second dataset whose cell the model covers, where none was expected. In Stage 11, [P1, P5, P6, after the read].

4. Discussion

4.1. Two clocks

The two neighbours disagree because they average differently. The lumped neighbour holds the cell's whole inventory at one temperature, so its own reaction heat adds to what it receives, and near its threshold it approaches 255.7 °C slowly: its crossing stretches from tens of seconds to minutes as

the delivered fraction falls toward the threshold (Fig. 6a). That induction is what lets it reproduce delays of 147 to 342 s with a few kilojoules delivered in 21.39 s. The hot spot sits on the line of contact and under the delivered flux, and its crossing is set by conduction, not by its own reaction: in Stage 2 the reaction heat moved its entry into the internal-short band by at most 0.01 s, and in Stage 11 its threshold did not move under any kinetic draw. It crosses within the delivery or not at all. A local criterion therefore answers whether the facing surface can be driven to the internal-short temperature, and the whole-cell clock how long the bulk takes once enough energy is in; neither alone is a model of the neighbour's runaway, which in the one published history began 271.5 s after the trigger's (Fig. 7a).

4.2. The delivered energy as the missing term

[comparators: the share of a failing cell's energy that leaves it, and what reaches a neighbour, with the numbers and conditions of each, beside this study's]

4.3. What sixteen tests cannot constrain

Each test reports one outcome, one time where it propagated and two maxima, from one thermocouple at an unreported angle. That is enough to show that no fixed fraction reproduces the outcomes (Section 3.4), but not enough to tell the models apart: held out, nothing the model yields beats one probability for every test (Section 3.7), and the one published history decides between the readings for its own test only. Four reports would change that. The neighbour's surface history, at two or more angles, separates a short delivery near threshold from a long low power, as PT_Exp4's did, and the angle decides whether the far face's overshoot is real (Section 3.5). The energy each trigger ejected, measured as fractional calorimetry measures it [9, 10], turns f^* and f_t from fitted fractions into tested predictions. And replicates at one trigger energy would measure the probability of propagation that the overlap of the two groups implies, which sixteen single tests at sixteen energies cannot.

4.4. What a pack model should do

A pack model that ignites a neighbour at a fixed delivered fraction will reproduce some tests and not others, because the fraction these tests need varies 42- to 208-fold. It should instead carry a distribution of delivered fraction, and report propagation as a probability; use the whole-cell clock for

when a neighbour runs away once it receives enough energy, and the local criterion for whether it can be driven there; and not read a time reproduced near a threshold as a validation, because under the kinetic draws the reproduced times move by up to [s] and the agreement within 10 s does not survive (Section 3.8).

4.5. Limits

One dataset carries the comparison: sixteen tests of one cell, in one rig. One history was digitised, from a figure, and its first 8.5 s are read as the thermocouple's exposure rather than the can. The delivery's place, shape and duration are choices, bracketed but not measured, and the trigger's release over 21.39 s rests on a 25 °C start: the later reading of the same figure ends the trigger's rise at 15.82 and 24.16 s, with PT_Exp4's trigger starting at 28.9 °C. The model has no sidewall rupture, no venting, no gas phase and no runaway of the neighbour as a whole, and its reaction heat is fed back one way. The kinetics were trained on NCM811 and graphite, not on the M50's silicon-graphite negative; the SEI is fresh or wholly aged; the emulation of the source's accelerating-rate test failed the acceptance set for it over the posterior and passed only as amended. The thermal properties are literature spreads, not measured uncertainties of this cell, and the bands of Stage 11 are over four draws a side. The module study, with more neighbours and more paths, is the next question.

5. Conclusions

[conclusions, written last]

Declarations

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Declaration of competing interest. The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability. The code, inputs, run records, checks and decisions of every stage are archived at Zenodo, `https://doi.org/10.5281/zenodo.23279455` (all versions; this paper uses version 1.7.0), and in the repository `https://github.com/siddharth161204-wq/m50-neighbour-onset-kinetics` [20]. The two-cell geometry is archived at `https://doi.org/10.5281/zenodo.23253738` [17].

CRediT authorship contribution statement. **Siddharth Satte:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work the author used Claude (Anthropic; model Claude Opus 5.5) in order to write and run the study's code, make and check its runs, read the published sources it uses, draft its decisions log, notes and figures, and draft this manuscript, each step under the author's direction; its use in the research is described in Section 2.10. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. [for the author's approval]

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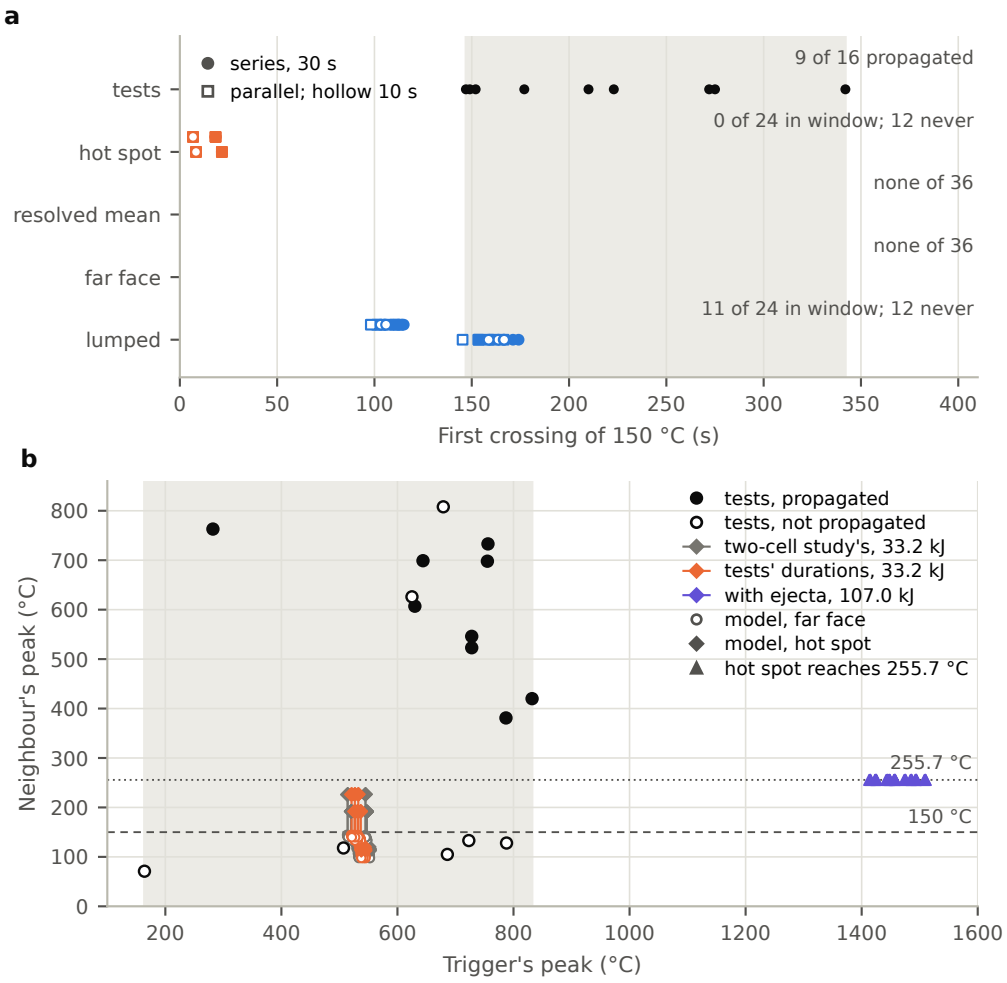


Figure 3: Conduction and radiation only, the cans in contact. (a) First crossing of 150 °C, the paper's criterion, by four clocks of the neighbour in 36 runs at the posterior-mean kinetics (three contacts, three tab conductances, releases of 10 and 30 s, both connections; within a row, the low, central and high contact from the bottom), against the nine measured times (top; shaded, their range). (b) The neighbour's peak against the trigger's: the sixteen tests (black) and the model's far face (circles) and hot spot (diamonds), joined per run, for the two-cell study's trigger, the trigger released over the tests' durations, and the Databank's total energy with the ejecta, 107.0 kJ; a run whose hot spot reaches 255.7 °C ends there (triangles). Shaded: the range of the measured trigger peaks.

		tests	C1 hot spot 255.7	C2 lumped 255.7	C3 lumped 150	C4 hot spot 138.5	C5 lumped 138.5	C6 hot spot 255.7 aged	C7 hot spot 138.5 aged
parallel	PT_Exp1 164 °C	no	no	no	no	no	no	no	no
	PT_Exp3 507 °C	no	no	no	182	15	138	no	15
	PT_Exp4 630 °C	**272 s**	no	no	108	13	92	no	13
	PP_Exp2 644 °C	**147 s**	no	no	104	13	88	no	13
	PT_Exp2 679 °C	no	no	no	95	13	81	no	13
	PP_Exp3 686 °C	no	no	no	93	12	80	no	12
	PP_Exp4 728 °C	**275 s**	21	no	84	12	73	21	12
	PP_Exp1 787 °C	**152 s**	18	277	74	11	65	18	11
series	ST_Exp3 282 °C	**177 s**	no	no	no	no	no	no	no
	SP_Exp4 625 °C	no	no	no	115	13	97	no	13
	ST_Exp2 723 °C	no	22	no	89	12	76	22	12
	SP_Exp1 728 °C	**223 s**	21	no	87	12	76	21	12
	SP_Exp2 755 °C	**149 s**	19	no	82	12	71	19	12
	ST_Exp4 756 °C	**342 s**	19	no	82	12	71	19	12
	ST_Exp1 788 °C	no	18	no	77	11	67	18	11
	SP_Exp3 832 °C	**210 s**	17	221	70	11	62	17	11
	hits	(9)	6	2	8	8	8	6	8
	misses		3	7	1	1	1	3	1
	false alarms		2	0	6	6	6	2	6
	correct rejections	(7)	5	7	1	1	1	5	1

Figure 4: The sixteen tests (rows; parallel above, series below, each in order of the trigger's measured peak) against the seven criteria at the central contact, each test with its own trigger: the time in seconds, from the start of the trigger's release, at which the criterion is met, or no. Shading: a hit or a correct rejection (blue tints), a false alarm (orange), a miss (pink). The first column gives the measured outcome and time; the bottom rows count the classes.

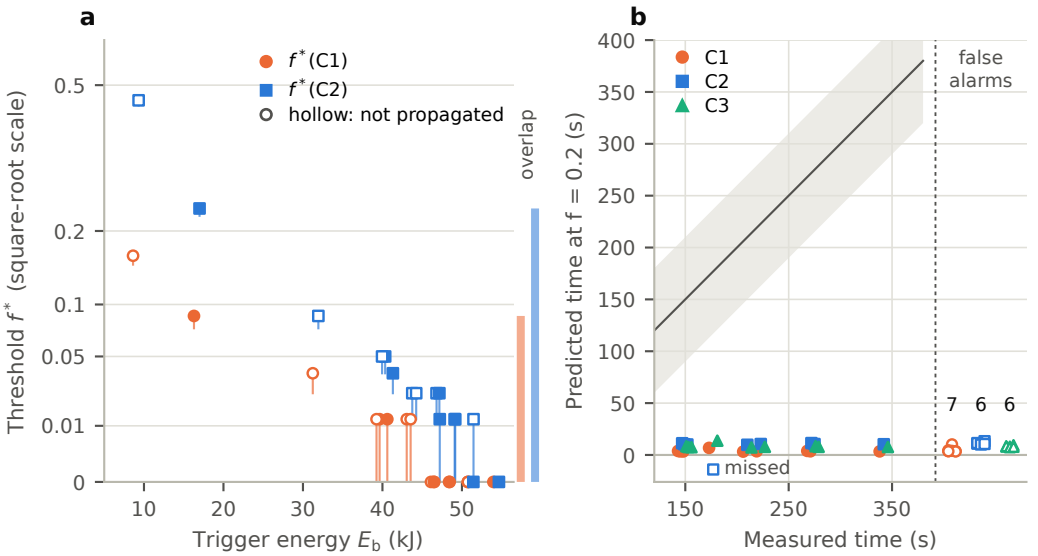


Figure 5: The delivered fraction the outcomes require. (a) f^*(C1) (circles) and f^*(C2) (squares) of the sixteen tests against the trigger's body energy, each as its bracket from the largest fraction that does not meet the criterion to the smallest that does (marker); filled, propagated; hollow, not. The bars at the right mark where the ranges of the two groups overlap. Square-root scale. (b) At a fraction common to every test, $f = 0.2$: each criterion's predicted time against the measured time for the propagated tests it predicts (a missed test below the axis) and, at the right, the predicted times of its false alarms, with their number. The line is equality, the band 60 s either side.

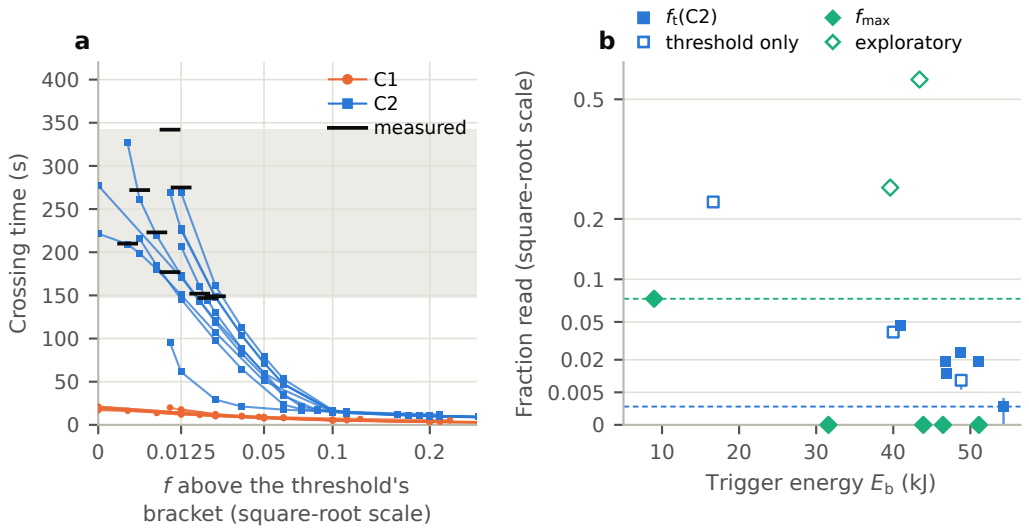


Figure 6: The fractions read from each test's own measurement. (a) With the 21.39 s delivery, each propagated test's crossing time against the fraction above the lower end of its threshold bracket, for the hot spot (C1) and the lumped neighbour (C2), one line per test and criterion; black bars, the measured times; shaded, their range. Square-root scale. (b) Against the trigger's energy: $f_{\mathrm{t}}(\mathrm{C2})$ for the propagated tests (hollow where the search ended at the threshold, shown with its bracket) and $f_{\max}$ at the far face for the others (hollow, the two exploratory maxima). Dashed: the largest registered $f_{\max}$ and the smallest $f_{\mathrm{t}}(\mathrm{C2})$.

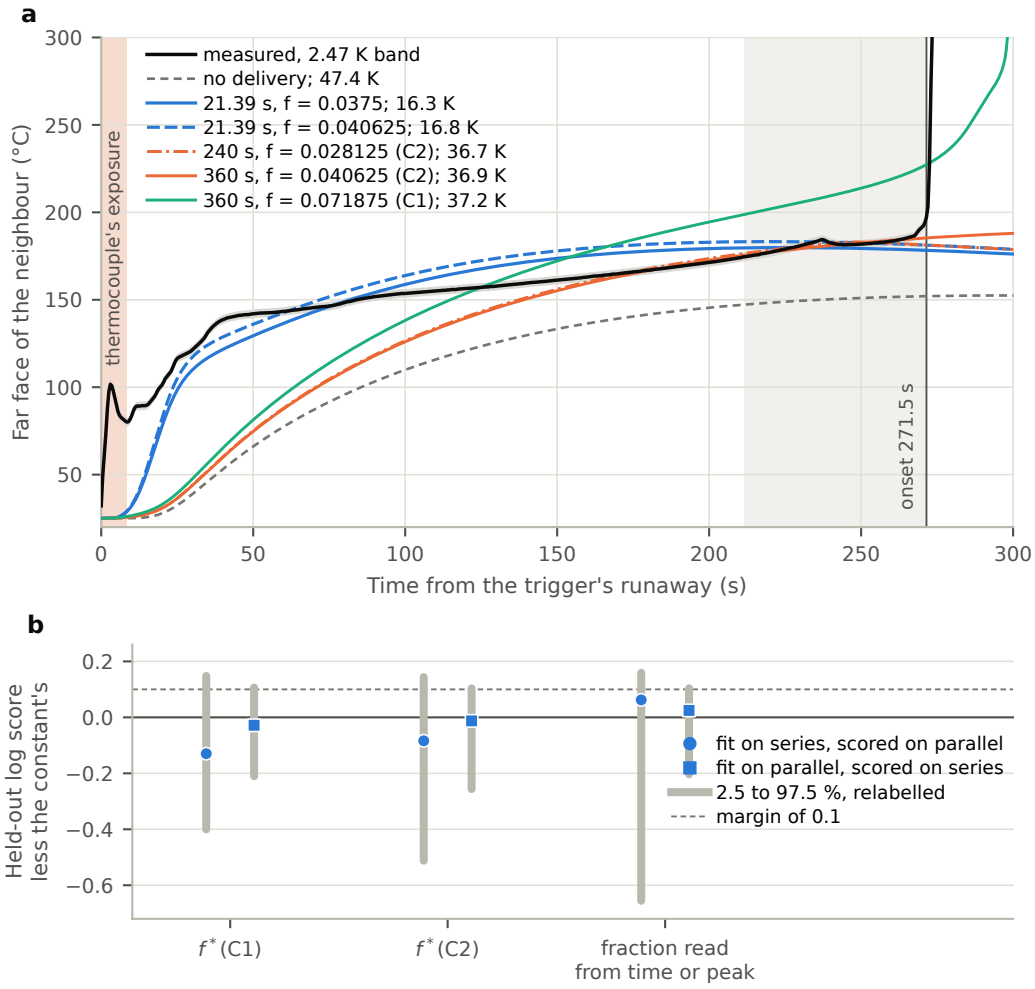


Figure 7: (a) PT_Exp4's measured thermocouple record (black; band, its 2.47 K reading uncertainty) and the model's far face at the central contact under the readings Stage 9 found for that test, each with its root mean square difference from the record up to the onset, 271.5 s (vertical line). Shaded at the left, the record's first 8.5 s, a spike read as the thermocouple's exposure at the trigger's runaway, not the can; at the right, the last 60 s before the onset. (b) Held out: the mean log score of the logistic fit in each predictor less that of one probability for every test, fitted on one connection and scored on the other (markers), with the 2.5 to 97.5 % band over 10,000 relabellings of the outcomes. Dashed: the registered margin of 0.1.